U.S. COMMERCIAL FISHERIES FOR MARLINS IN THE NORTH PACIFIC OCEAN[1]

Russell Y. Ito and Michelle Sculley
Pacific Islands Fisheries Science Center
NOAA Fisheries Service
Honolulu, Hawaii 96818 U.S.A.

November 2, 2020

[1] PIFSC Working Paper WP-20-___
Issued __-___-____

Working document submitted to Intersessional Meeting of the Marlin Working Group, International Scientific Committee for Tuna and Tuna-like Species in the North Pacific Ocean, November 2-12 2020, Honolulu, Hawaii, U.S.A. Document not to be cited without author's permission.

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INTRODUCTION

This report summarizes historical trends and recent developments for U.S. commercial fisheries taking marlins and related billfish species (Istiophoridae) in the North Pacific Ocean. Five species of marlins are caught by U.S. commercial fisheries in the North Pacific Ocean. These are striped marlin (*Kajikia audax*), blue marlin (*Makaira nigricans*), shortbill spearfish (*Tetrapturus angustirostris*), sailfish (*Istiophorus platypterus*), and black marlin (*Istiompax indica*). The first two species are predominant in the commercial landings. The description of fisheries in this report will serve as background information for stock assessment and standardization models developed in the ISC Billfish Working Group.

1. FISHERIES AND CATCHES

U.S. fisheries for marlins in the North Pacific Ocean can be categorized according to three distinct gear types: longline, troll, and handline. The U.S. longline fishery is the largest and includes Hawaii and California-based vessels (Table 1). This fishery catches marlins incidentally on sets targeting tuna or swordfish. Troll fisheries in Hawaii, Guam, and the Commonwealth of the Northern Mariana Islands (CNMI) take the second largest catches of marlins. These fisheries opportunistically target marlins on a seasonal basis. The Hawaii handline fishery represents the third category, with small incidental catches of marlin. Blue marlin landings from both longline and troll fisheries (Fig. 1) were typically the largest component of the marlin landings (Table 2), followed by striped marlin, landed primarily by the longline fishery (Fig. 2), and shortbill spearfish ranking third. Marlins are also caught by recreational fisheries but there is no mandatory data collection program for this fishery sector, therefore, only the U.S. commercial fisheries are discussed in this report.

U.S. Longline Fisheries

The longline gear consists of a single monofilament mainline about 30 to 80 km in length with floats attached to the mainline to support the gear in the water column. Branchlines with baited hooks are attached to the mainline between the floats. Gear configurations and operational techniques differ according to target species (i.e., tunas, *Thunnus* spp., and swordfish, *Xiphias gladius*) (Ito and Machado 2001). Deep-set longline fishing targets tunas. Gear is usually set in the morning and then hauled back around sunset; Pacific sauries *Cololabis saira* or sardines (Clupeidae) are used for bait; 15 or more hooks are set between floats; and a line thrower is used (Kawamoto et al. 1989). The latter creates slack in the mainline, which causes the gear to sag between floats as it sinks and results in a "deep-set". In contrast, shallow-set longline fishing is used to target swordfish and gear is typically set after dusk and with haulback beginning around dawn the following morning, uses mackerel *Scomber japonicus* or mackerel-like bait, attaches chemical lightsticks to the branchlines, and typically sets 4-5 hooks between floats (Ito et al 1994, PIRO 2014). Since the gear and technique for swordfish is fishing relatively shallow, a line thrower is not needed and is referred to as "shallow-set" longline fishing. Striped and blue marlin were the largest components of the longline marlin landings followed by shortbill spearfish (Table 3). The deep-set longline sector accounts for majority of the effort and marlin landings for this fishery.

The Hawaii-based vessels in the U.S. longline fishery have operated under a limited-access program since 1994. This program capped participation at 164 vessels, although the number of active vessels has never reached this limit. Participation by the U.S. longline fishery peaked at 150 active vessels in 2019 (deep- and shallow-set sectors combined).

Two other important characteristics of this fishery are its geographic range and the annual number of hooks set. The U.S. longline fishery ranged from 10o N to 35^{o} N latitude and from 120^{o} W to 175^{o}W longitude in 2019. The total range exploited since 1991 extended from 5^{o}S to 50^{o}N latitude and from 125^{o}W to 175^{o}E longitude. Effort by the U.S. longline fishery was a record 63.5 million hooks set in 2019. The record effort was due to more hooks set by the deep-set sector of the longline fishery which accounted for 99% of the total number of hooks set and an expansion of longline fishing outside of the U.S. EEZ in 2019.

Longline catch of striped marlin varied from 1987 with peaks at more than 660 mt in 1991 and 1995 and lows of 167 mt in 2002 and 2010. Striped marlin catches were on a increasing trend from 2011 with catch at 548 mt in 2019. Longline catch of blue marlin increased rapidly from 37 mt 1987 to 378 mt in 1990 and remained stable through 1997. Blue marlin catches then declined to 197 mt in 1997 and was on an increasing trend peaking at 687 mt in 2019.

The ISC Billfish Working Group concluded there were two stocks of striped marlin in the North Pacific Ocean (ISC 2010) and the boundary between the Western-central Pacific Ocean (WCPO) and Eastern Pacific Ocean (EPO) stocks was delineated at 150^{o} W. U.S. longline catch of striped marlin by stock boundary show 81% of the catch originating from the area at or west of 150^{o} W during 2010-2019 (Table 4). Striped marlin caught east of 150^{o} W and undisclosed areas were highest in 2017 and represented 39% of the catch by the longline fishery. The deep-set sector of the longline fishery was responsible for 97% of the striped marlin catch during 2010-2019.

Plots of the geographic distribution of striped marlin catches (number of fish) show that the highest catches occurred around the the main Hawaiian Islands (15°N-30°N and 155°W-165°W) while the highest blue marlin catches occurred south-southwest of the main Hawaiian Islands (15°N-20°N and 155°W-165°W) (Figs. 3 & 4). Catches of marlins exhibited strong seasonal cycles. Striped marlin catches were typically highest in the first and fourth quarters of the year, whereas blue marlin catches were usually highest in the second and third quarters of the year.

Nominal catch per unit effort (CPUE: number of fish per 1000 hooks) for the two marlin species exhibited declines from the early 1990s into the early 2000s. Striped marlin CPUE on deep-set longline fishing peaked at 2.2 fish per 1000 hooks in 1992, exhibited a significant decline through 2000, and remained low thereafter. CPUE continued to decrease further to a record low of 0.10 in 2010 and was slightly higher at 0.24 in 2019 (Fig. 5). Blue marlin deep-set longline CPUE exhibited a peak of 0.68 in 1991, decreased sharply in 1992, declined slowly to a record low 0.07 in 2012, and increased to 0.20 in 2019.

The weight-frequency histogram for striped marlin caught by the U.S. longline fishery, derived from records of commercial fish landings (see data sources below), had a strong mode in the 11-15 kg and 16-20 kg weight intervals in 2019 (Fig. 6A). The mean weight for striped marlin was 26.4 kg. The blue marlin weight-frequency distribution was unimodal with a peak at the 51-60 kg weight interval in 2019 (Fig. 6B). The mean weight for blue marlin was 72.1 kg.

Hawaii, Guam, and CNMI Troll Fisheries

The troll fisheries in Hawaii, Guam, and CNMI are hook and line fisheries that use fishing gear consisting of fiberglass rods, reels and artificial lures typically made of resin or chrome metal heads dressed with colored rubber skirts (Rizzuto 1977). Live bait bridled to hooks are also used to catch marlins and other pelagic fishes. This fishery targets tunas, marlins and other pelagic species such as mahimahi (*Coryphaena spp.)* and wahoo *(Acanthocybium solandri*). Fishing is conducted from relatively small boats, e.g. <15 m.

The number of troll fishers ranged from 1,704 in 1988 to 2,367 in 1999 with 1,805 fishers in 2019. Seventy-one percent of the troll fishers were from Hawaii, 26% from Guam and 3% from CNMI in 2019. The duration of a typical troll trip is one day. Since this fishery employs small vessels, most trips remain within 100 km from shore, well inside the 200 mile U.S. EEZ.

Blue marlin was the predominant component of troll marlin catch. Blue marlin catch peaked at 434 mt in 1996, declined to a record low 128 mt in 2007, and was 178 mt in 2019 (Table 5). Striped marlin and other marlin species represented only a small proportion of the marlin catch at 13 mt in 2019. Blue marlin and striped marlin catch were higher in the earlier years of the time series.

Marlin CPUE for the Hawaii troll fishery was expressed as kg of fish per day. Blue marlin CPUE was higher than striped marlin CPUE, Blue marlin CPUE exhibited a increasing trend from 2012 while striped marlin CPUE remained low (Fig. 7)

Hawaii Handline Fishery

The Hawaii handline fishery, which targets tunas, includes day and night components known as the "palu ahi" and "ika shibi" fisheries, respectively. The daytime handline fishery employs "palu" (chum in Hawaiian) to evoke a feeding frenzy in an aggregation of juvenile "ahi" (tuna in Hawaiian) and hook the catch with a handline. The nighttime handline fishery has two sets of gear, one used to catch the "ika" (squid in Japanese) for bait and the other for catching large "shibi" (tuna in Japanese) (Yuen 1979).

There were 444 handline fishers in 2019. The duration of a handline trip is typically one day for the daytime handline fishery and one night for the nighttime handline fishery. As with the troll fisheries, most handline trips remain within 100 km from shore, although some handline fishers operate offshore by seamounts and weather buoys on multiple day trips.

Marlins are rarely caught by the handline fishery and represent only a small proportion of its overall catch. This fishery caught relatively small amounts of striped and blue marlins when compared to the longline and troll fisheries. There has been no striped marlin catch by the handline fishery from 2005. The highest striped marlin catch was 2 mt in 2001 (Table 6). Handline catch of blue marlin were higher in the earlier years, peaked at 9 mt in 1997 and was 5 mt in 2019.

The weight-frequency histogram for striped marlin caught on troll and handline gear exhibited a peak at the 16-20 kg interval (Figure 8A). The mean weight for striped marlin was 31.3 kg in 2019. The blue marlin weight-frequency distribution was unimodal with a mean weight of 72.1 kg in 2019 (Figure 8B). The weight-frequency distribution of blue marlin by the troll and handline fisheries was nearly identical to the histogram for the longline fishery.

2. DATA SOURCES

Category I: Annual Catch Data

Category I catch statistics refer only to the quantity of fish kept and landed. Catch that was discarded or released was not included. Several sources of fisheries dependent data for the longline, troll, and handline fisheries are collected by Federal (NOAA Fisheries Service), State (Hawaii), and Pacific Island (Guam and CNMI) agencies and used in combination by staff of the NOAA Pacific Islands Fisheries Science Center (PIFSC). The duration and coverage (i.e., percent of catch reported) varied amongst the different data sources (Table 7).

Estimated catches are reported in this paper as whole weights. Some fish were landed whole while others were processed at sea, e.g., headed and gutted or gilled and gutted. The recorded weight of individual processed fish was adjusted by applying a conversion factor depending on the degree of processing (Table 8). This step increased the nominal weight of processed catch to a whole weight estimate to account for the weight loss. Likewise, to account for missing market sample days, sample data were extrapolated to represent full coverage to estimate total landings.

Data sets were combined to estimate annual catch statistics for certain fisheries. For example, the U.S. longline fishery catch was estimated from Federal logbook data, market sample data, and State of Hawaii, Division of Aquatic Resources (Hawaii DAR) Commercial Marine Dealer data. The numbers of fish kept, as recorded in longline logbooks, are multiplied by the mean weights of landed fish, estimated from the PIFSC market sample data or the Hawaii DAR Commercial Marine Dealer data.

Marlin Species Identification Issues

Since blue marlin, striped marlin, and black marlin are similar in appearance, a longstanding problem in monitoring the Hawaii-based longline fishery at the NOAA PIFSC has been the accuracy of species identifications for the istiophorid billfishes. This problem has primarily affected logbook data, but some fishery observers, particularly newly-hired individuals, have also erred in species identifications. A long-term project to correct these problems was completed for the years 1995 through 2003. Its principal output consisted of one paper emphasizing blue marlin that was published in a peer-reviewed scientific journal that addressed the five istiophorid species (Walsh et al. 2005). A subsequent document showed the overall marlin counts in the Hawaii-based longline logbook data were reasonably accurate but blue marlin was overlogged by 18% while striped marlin was underlogged 11% during the study period (Walsh, W.A. et al. 2007)(Figure 9). The document can be obtained from the PIFSC's website at:

http://www.pifsc.noaa.gov/library/pubs/tech/NOAA_Tech_Memo_PIFSC_13.pdf

The marlin species identification corrections from the project are included in the longline marlin catch tables for the years of these studies. Nominal marlin catches were reported for the years prior to 1995 and after 2003. This Working Paper has been written to conform to the guidelines adopted by the ISC concerning use of best available scientific information (Brodziak and Dreyfus 2011). The specific guidelines pertaining to this Working Paper are related to the need for accurate species identifications.

Category II: Spatial Catch and Effort Data

Year, area fished, catches and effort are the most important information included in Category II data reporting. The U.S. longline fishery provided Category II data calculated from Federal logbook and Hawaii DAR Commercial Marine Dealer data. The combination of data sets was sufficient to generate area-specific summaries of catch and effort.

Category III: Biological (size composition) Data

Biological measurements were obtained for the Hawaii longline, troll, and handline fisheries. Raising factors were applied to the market sample and Hawaii DAR Commercial Marine Dealer data if the fish was processed to yield an estimated whole weight (Table 7). Weight-frequency distributions for striped marlin and blue marlin were produced from HDAR Commercial Marine Dealer data.

The Pacific Islands Fishery Observer Program (PIROP) provides detailed set-by-set data on the Hawaii-based longline fishery including the eye-fork length of fish caught in cm and a variety of operational variables, among them: location as latitude and longitude, vessel ID, hooks per float, and total number of hooks set, among others. Data are collected following the procedures outline in the PIROP observer manual (Pacific Islands Regional Office, 2017). Data were extracted from the PIROP database on 15 October 2020.

Blue marlin are caught as non-target species in both the deep-set tuna-targeting and shallow-set swordfish-targeting sectors of the Hawaii-based longline fleet. Observers were first placed onboard longline vessels in 1994. Observer coverage varied significantly prior to 2000, with observer coverage between 3.3 and 10.4% each year for the entire fishery (NMFS, 2017). Due to interactions with protected species the shallow-set sector was closed from 2001 to2004 and when it was reopened, 100% observer coverage occurred on shallow-set trips and ~20% observer coverage occurred on deep-set trips (Gilman et al., 2007). The deep-set trips are typically further south than the shallow-set trips, which are concentrated around the sub-tropical frontal zone (STFZ) where large swordfish are caught (Bigelow et al., 1999). Here, deep-sets are defined as sets with greater than 10 hooks per float prior to 2004 and greater than 14 hooks per float after 2004. This has been shown to better identify the recorded targeted fish by fishers than using greater than 14 hooks per float for all years (Sculley, 2019).

Prior to 2006 observers measured every fish caught. Since 2006, observers measured every third fish caught, regardless of species. Spatial plots are in 5x5 degree

squares where cells with fewer than three vessels measuring blue marlin are excluded for confidentiality. Figures were made in R (version 3.4.0, R Core Team, 2017).
The mean eye-fork length of blue marlin caught in the Hawaii-based longline fishery was 165 cm. The size of blue marlin caught each year varied slightly between 160 and 180 cm EFL for most of the years in the time series (Figure 10). A single peak in length for blue marlin at around 165 cm EFL (Figure 11), and there does not appear to be substantial differences in mean length by quarter (Figure 12). There does not appear to be significant differences in the length of blue marlin caught in the deep- and shallow-set sectors of the longline fleet (Figure 13) nor between male and female blue marlin, although the size differences do not tend to be as large at the smaller sizes caught by the Hawaiian longline fleet (Figure 14).

The spatial patterns of fish size by length are illustrated in Figure 15 and Figure 16. The largest fish are caught north and east of the Hawaiian Islands while those caught to the south and west are smaller. Overall, fish tend to be larger in quarters two and three. Overall, it appears that the Hawaii-based longline fishery catches primarily small blue marlin as non-target species. Mean eye-fork length has been relatively consistent throughout the time series. Larger fish are typically caught to the north of Hawaii and smaller fish are typically caught to the south of Hawaii.

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Table 1.—Annual U.S. commercial fisheries marlin catch* (mt) from the North Pacific Ocean by gear type, 1987-2019.

	Gear type			
Year	Longline	Troll	Handline	Total
1987	368	324	9	701
1988	675	362	7	1,044
1989	1,100	404	6	1,510
1990	973	373	6	1,352
1991	1,029	444	6	1,479
1992	947	351	5	1,303
1993	910	422	6	1,338
1994	787	385	4	1,176
1995	1,179	424	5	1,608
1996	884	504	8	1,396
1997	944	467	10	1,421
1998	831	305	3	1,139
1999	822	387	6	1,215
2000	464	269	3	736
2001	666	368	4	1,038
2002	367	269	3	639
2003	812	255	4	1,071
2004	868	243	4	1,115
2005	1,069	220	3	1,292
2006	1,187	193	3	1,383
2007	698	153	1	852
2008	1,016	208	1	1,225
2009	745	197	1	943
2010	615	179	2	796
2011	1,001	233	2	1,236
2012	763	164	2	929
2013	1,029	160	3	1,192
2014	1,197	184	4	1,385
2015	1,396	224	3	1,623
2016	1,312	193	2	1,507
2017	1,412	173	4	1,589
2018	1,360	192	3	1,555
2019	1,666	202	5	1,873

* Based on estimated whole weight.

Table 2.--Annual U.S. commercial fisheries marlin catch* (mt) from the North Pacific Ocean by species, 1987-2019.

Year	Striped marlin	Blue marlin	Shortbill spearfish	Other marlins	Total
1987	303	334	43	21	701
1988	559	398	65	22	1,044
1989	636	721	128	25	1,510
1990	565	715	50	22	1,352
1991	703	684	60	32	1,479
1992	498	648	46	111	1,303
1993	540	678	54	66	1,338
1994	360	696	59	61	1,176
1995	716	758	65	69	1,608
1996	513	804	38	41	1,396
1997	463	851	47	60	1,421
1998	479	541	63	56	1,139
1999	443	616	96	60	1,215
2000	224	418	43	51	736
2001	427	521	40	50	1,038
2002	197	368	39	35	639
2003	547	409	80	35	1,071
2004	420	471	186	38	1,115
2005	536	525	207	24	1,292
2006	625	570	161	27	1,383
2007	291	390	147	24	852
2008	443	529	226	27	1,225
2009	271	540	113	19	943
2010	186	467	119	24	796
2011	381	587	236	32	1,236
2012	292	452	163	22	929
2013	405	545	213	29	1,192
2014	437	698	218	31	1,385
2015	501	828	263	32	1,623
2016	402	725	339	40	1,507
2017	412	845	303	29	1,589
2018	477	831	219	29	1,555
2019	561	1,085	199	29	1,873

* Based on estimated whole weight.

Table 3.—U.S. longline commercial marlin catch* (mt) from the North Pacific Ocean, 1987-2019.

Year	Striped marlin	Blue marlin	Shortbill spearfish	Other marlins	Total
1987	272	51	43	2	368
1988	504	102	65	4	675
1989	612	356	128	4	1,100
1990	538	378	50	7	973
1991	663	297	60	9	1,029
1992	459	347	46	95	947
1993	471	339	54	46	910
1994	326	362	59	40	787
1995	664	407	65	43	1,179
1996	458	363	38	25	884
1997	424	429	47	44	944
1998	453	277	63	38	831
1999	414	284	96	28	822
2000	209	183	43	29	464
2001	383	227	40	16	666
2002	167	137	39	24	367
2003	517	198	80	17	812
2004	385	283	186	14	868
2005	516	337	207	9	1,069
2006	604	409	161	13	1,187
2007	278	261	147	12	698
2008	429	348	226	13	1,016
2009	261	360	113	11	745
2010	167	317	119	12	615
2011	365	386	236	14	1,001
2012	281	309	163	10	763
2013	397	406	213	13	1,029
2014	425	534	218	19	1,197
2015	490	629	263	15	1,396
2016	390	562	339	20	1,312
2017	406	687	303	16	1,412
2018	465	664	219	13	1,360
2019	548	902	199	18	1,666

* Based on estimated whole weight.

Table 4.—U.S. deep- and shallow-set longline commercial striped marlin catch (mt) by stock boundary in the North Pacific Ocean, 2010-2019.

	Deep-set				Shallow-set				Fleet			
Year	West of 150^{o} W	East of 150^{o} W	No longitude	Total	West of 150o W	East of 150o W	No longitude	Total	West of 150o W	East of 150o W	No longitude	Total
2010	120	25	10	155	8	2	2	12	128	26	13	167
2011	305	28	12	345	14	2	4	20	319	29	17	365
2012	244	15	10	270	11	0	1	12	255	16	11	281
2013	302	66	13	381	13	0	3	16	315	66	16	397
2014	331	65	15	411	10	0	4	14	340	66	19	425
2015	387	76	16	479	7	1	3	11	394	77	19	490
2016	301	59	17	377	8	1	5	13	309	59	22	390
2017	305	72	15	391	12	0	4	16	316	72	18	406
2018	366	85	12	463	1	0	1	2	367	85	13	465
2019	445	84	19	547	0	0	0	1	445	84	19	548
Mean	310.5	57.4	14.0	382.0	8.2	0.6	2.7	11.5	318.7	58.0	16.7	393.5

Table 5.—U.S. troll fishery marlin catch* (mt) from the North Pacific Ocean, 1987-2019.

Year	Striped marlin	Blue marlin	Shortbill spearfish	Other marlins	Total
1987	30	275	0	19	324
1988	54	290	0	18	362
1989	24	359	0	21	404
1990	27	331	0	15	373
1991	40	381	0	23	444
1992	38	297	0	16	351
1993	68	334	0	20	422
1994	34	330	0	21	385
1995	52	346	0	26	424
1996	54	434	0	16	504
1997	38	413	0	16	467
1998	26	261	0	18	305
1999	28	327	0	32	387
2000	14	233	0	22	269
2001	42	292	0	34	368
2002	30	228	0	11	269
2003	29	208	0	18	255
2004	34	186	0	23	243
2005	20	185	0	15	220
2006	21	158	0	14	193
2007	13	128	0	12	153
2008	14	180	0	14	208
2009	10	179	0	8	197
2010	19	148	0	12	179
2011	16	199	0	18	233
2012	11	141	0	12	164
2013	8	136	0	16	160
2014	12	160	0	12	184
2015	11	196	0	17	224
2016	12	161	0	20	193
2017	6	154	0	13	173
2018	12	164	0	16	192
2019	13	178	0	11	202

* Based on estimated whole weight.

Table 6.—The U.S. handline fishery marlin catch* (mt) from the North Pacific Ocean, 1987-2019.

Year	Striped marlin	Blue marlin	Shortbill spearfish	Other marlins	Total catch
1987	1	8	0	0	9
1988	1	6	0	0	7
1989	0	6	0	0	6
1990	0	6	0	0	6
1991	0	6	0	0	6
1992	1	4	0	0	5
1993	1	5	0	0	6
1994	0	4	0	0	4
1995	0	5	0	0	5
1996	1	7	0	0	8
1997	1	9	0	0	10
1998	0	3	0	0	3
1999	1	5	0	0	6
2000	1	2	0	0	3
2001	2	2	0	0	4
2002	0	3	0	0	3
2003	1	3	0	0	4
2004	1	2	0	1	4
2005	0	3	0	0	3
2006	0	3	0	0	3
2007	0	1	0	0	1
2008	0	1	0	0	1
2009	0	1	0	0	1
2010	0	2	0	0	2
2011	0	2	0	0	2
2012	0	2	0	0	2
2013	0	3	0	0	3
2014	0	4	0	0	4
2015	0	3	0	0	3
2016	0	2	0	0	2
2017	0	4	0	0	4
2018	0	3	0	0	3
2019	0	5	0	0	5

* Based on estimated whole weight.

Table 7.—Data sources and rates of coverage for the longline, troll, and handline fisheries by category.

	Hawaii-based longline	Hawaii troll	Guam troll	CNMI troll	Hawaii handline
Category I: Annual catch data					
Market sample	~33-90%	+++	---	---	+++
Fish dealer	~50-100%	+++	---	+++	+++
Logbook	~100%	---	---	---	---
Fish catch report	---	+++	---	---	+++
Creel survey	---	---	+++	---	---
Observer	NA	NA	NA	NA	NA
Category II: Spatial catch and effort data					
Market sample	NA	NA	NA	NA	NA
Fish dealer	NA	NA	NA	NA	NA
Logbook	~100%	---	---	---	---
Fish catch report	---	+++	---	---	+++
Creel survey	NA	NA	NA	NA	NA
Observer					
Category III: Biological (size composition) data					
Market sample	~33-90%	+++	---	---	+++
Fish dealer	~50-100%	+++	---	+++	+++
Logbook	NA	NA	NA	NA	NA
Fish catch report	NA	NA	NA	NA	NA
Creel survey	---	---	+++	---	---
Observer	3-25%	---	---	---	---

*NA - not applicable, +++ - available but coverage unknown, --- - not collected

Table 8.—Conversion factors for processed fish.

Species	Condition of fish	Raising factor
Blue marlin	Shark bitten	1.11
	Gutted	1.15
	Gilled & gutted	1.25
	No head	1.28
	No head & guts	1.47
	No head, guts & tail	1.54
Striped marlin	Shark bitten	1.11
	Gutted	1.15
	Gilled & gutted	1.23
	No head	1.25
	No head & guts	1.37
	No head, guts & tail	1.41

Figure 1.—Catch (mt) of blue marlin by U.S. commercial fisheries in the North Pacific Ocean, 1987-2019.

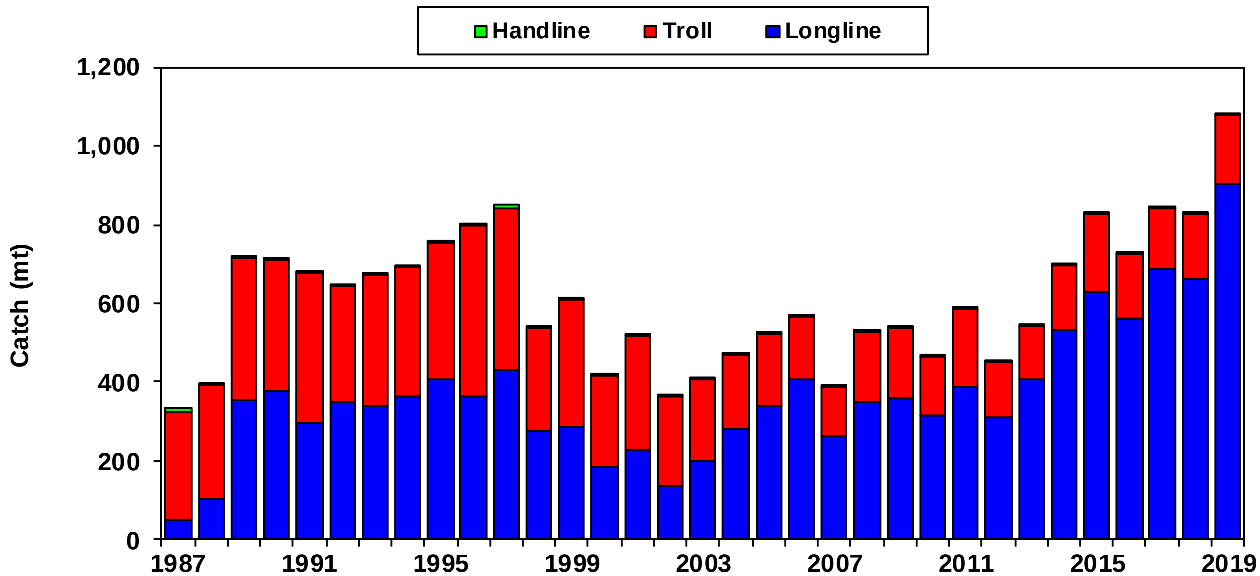


Figure 2.–Catch (mt) of striped marlin by U.S. commercial fisheries in the North Pacific Ocean, 1987-2019.

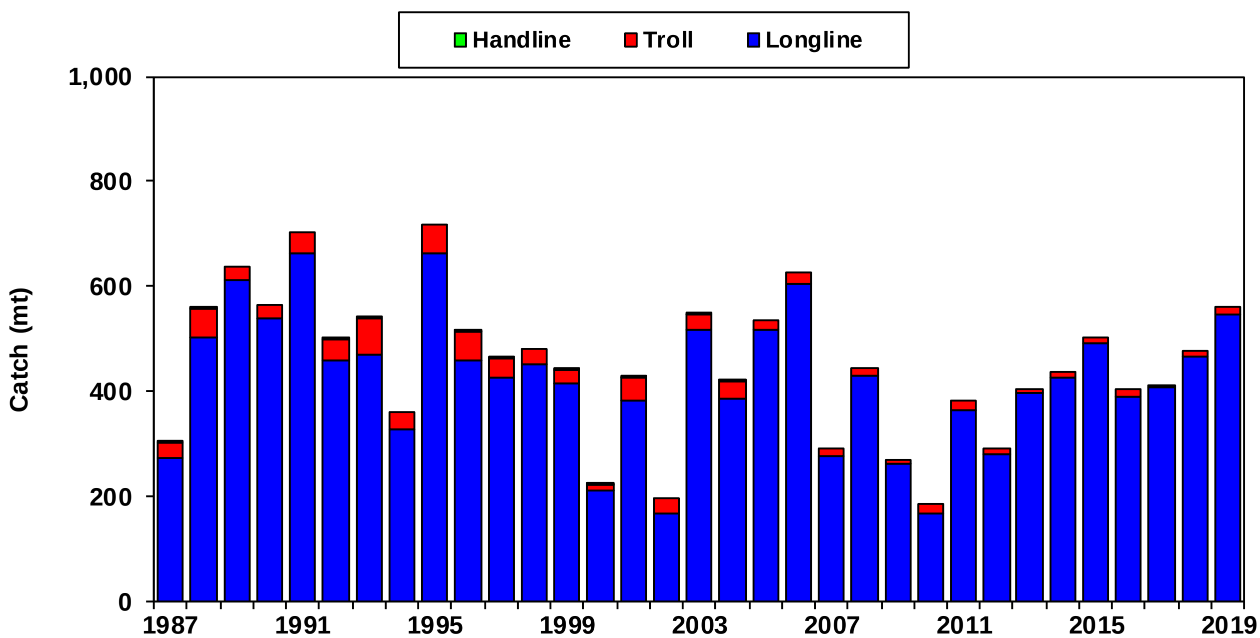


Figure 3.—U.S. longline striped marlin catch (number of fish) by area, 2019.

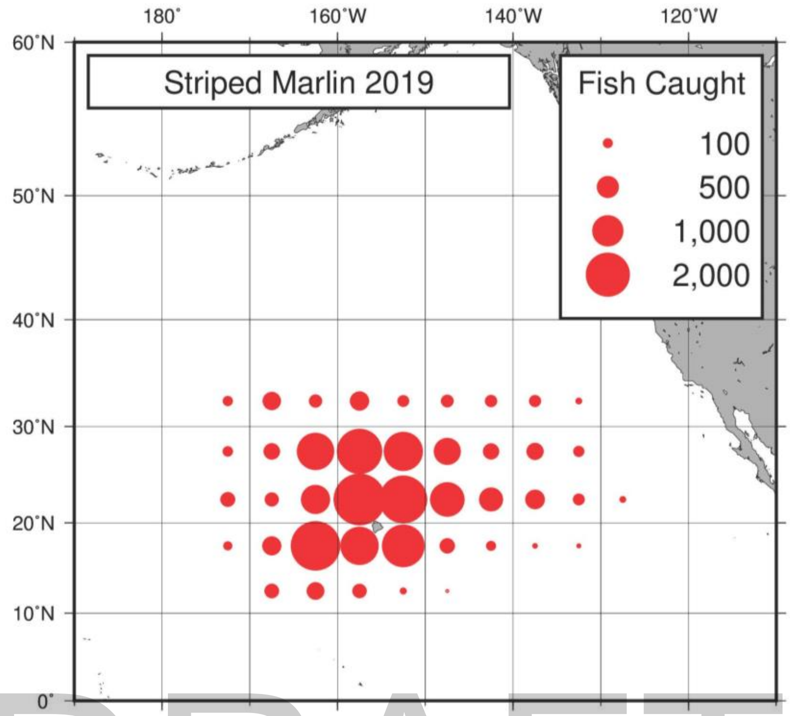


Figure 4.—U.S. longline blue marlin catch (number of fish) by area, 2019.

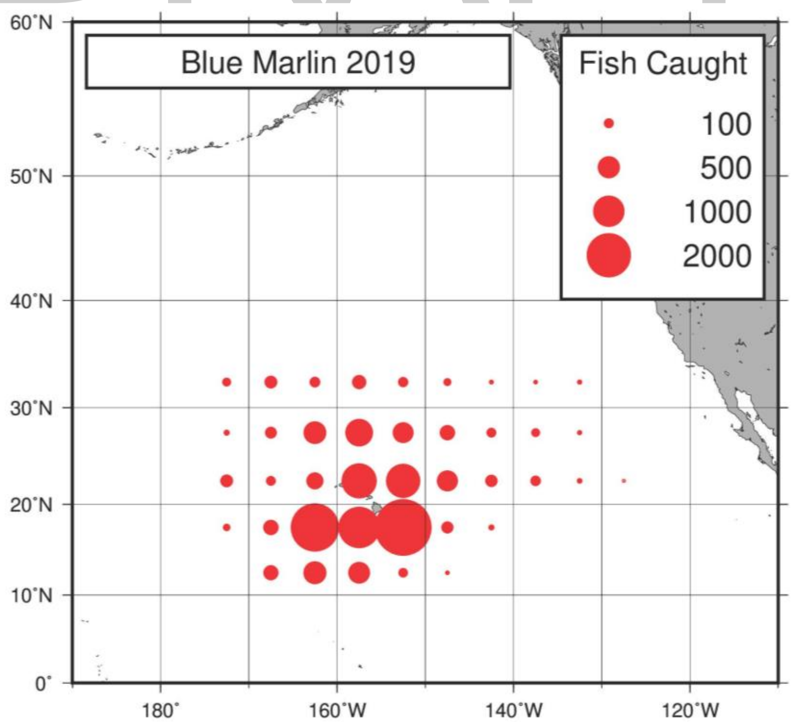


Figure 5.—U.S. longline striped marlin and blue marlin nominal CPUE on tuna-targeted deep sets, 1991-2019.

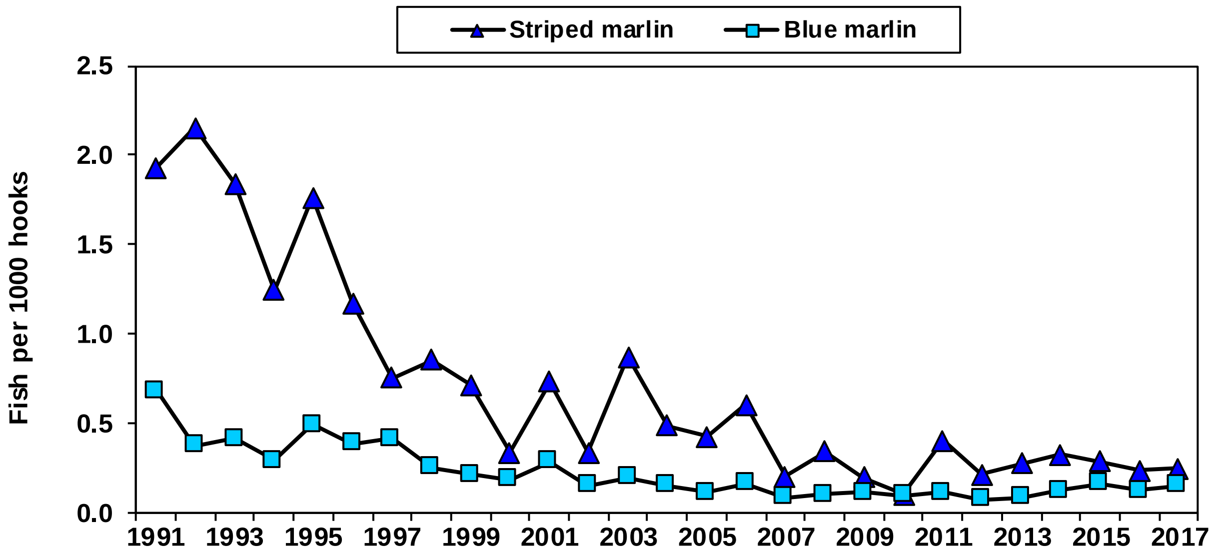
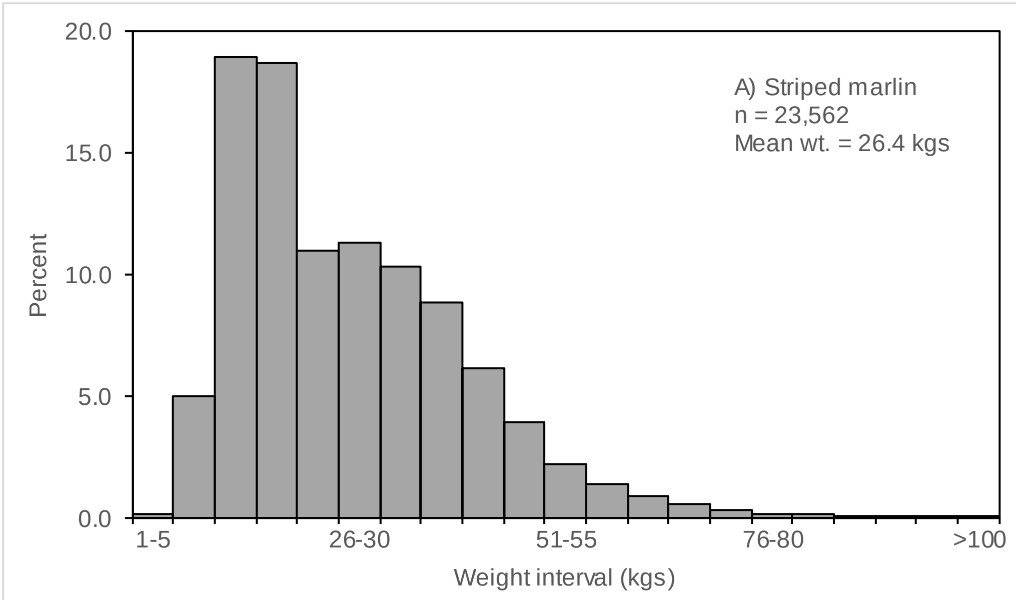


Figure 6.—U.S. longline A) striped marlin and B) blue marlin weight-frequencies, 2017.



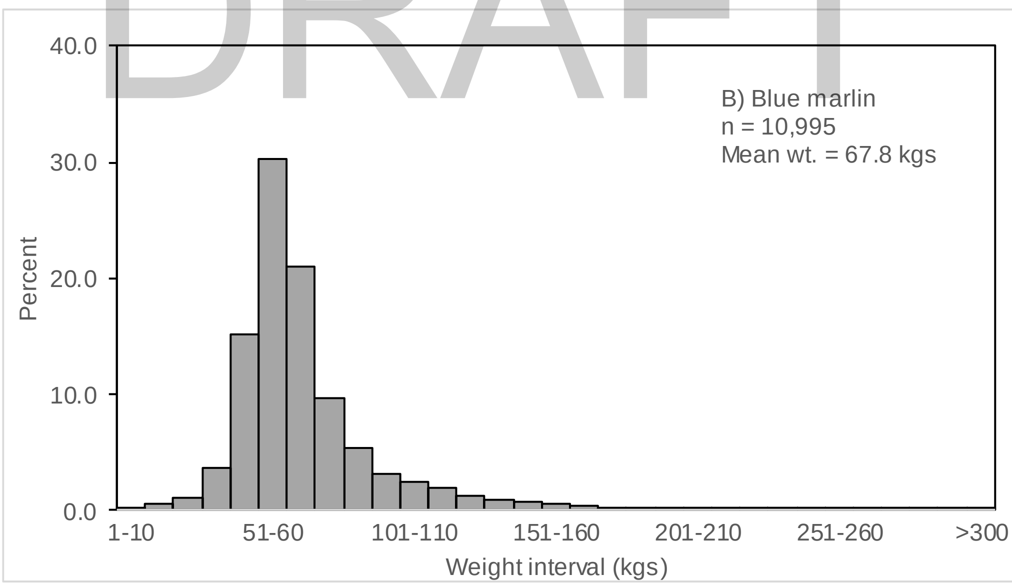


Figure 7.—Hawaii troll striped marlin and blue marlin nominal CPUE, 1991-2019.

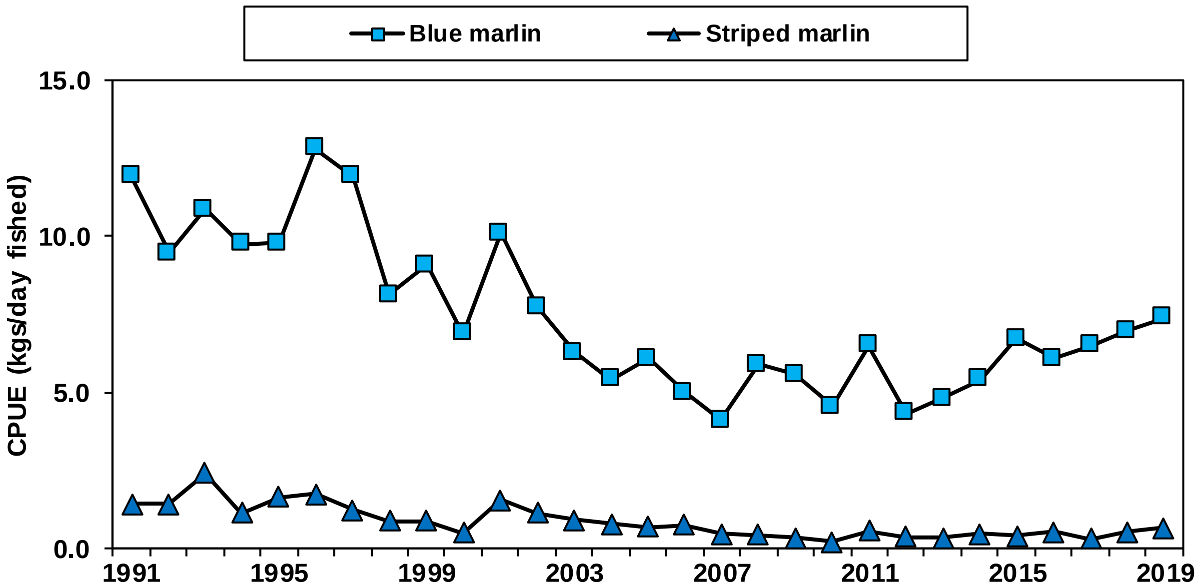
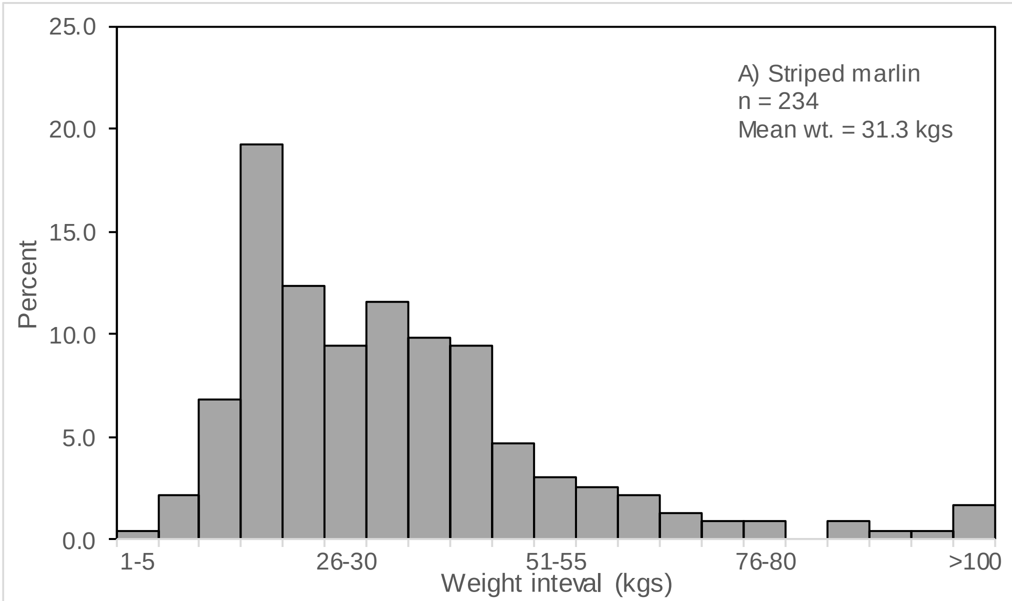


Figure 8.—Hawaii troll and handline A) striped marlin and B) blue marlin weight-frequencies, 2019.



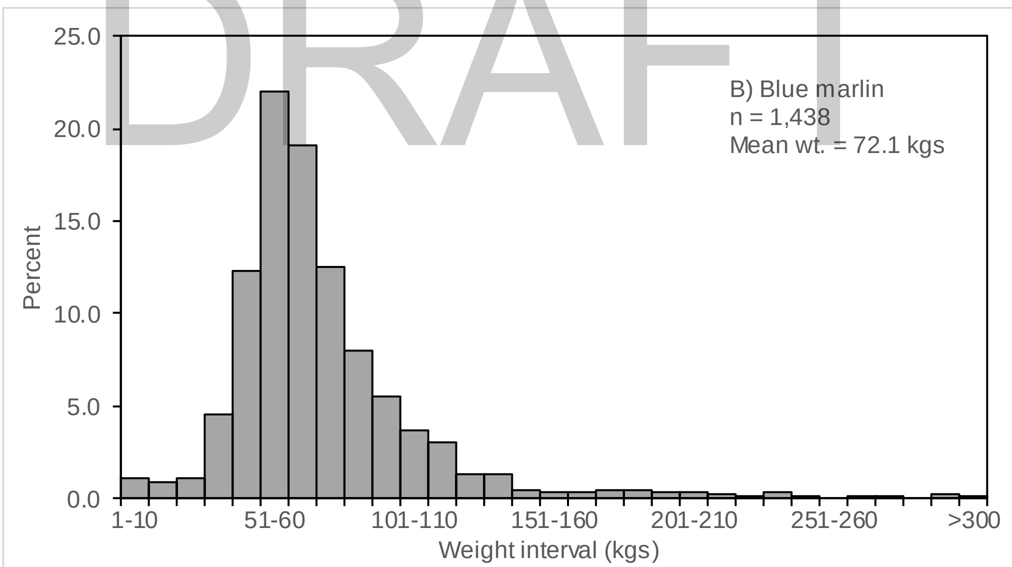
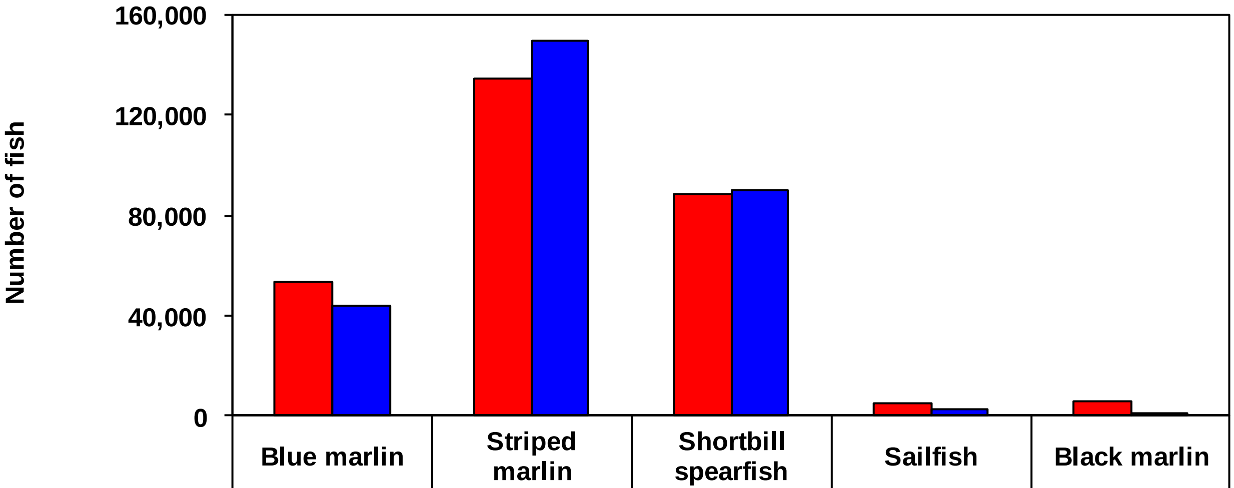


Figure 9. Nominal and corrected marlin catches for the Hawaii-based longline fishery, 1995-2003. Source: Walsh et al. 2007, Table B1.



	Blue marlin	Striped marlin	Shortbill spearfish	Sailfish	Black marlin
Nominal catch	53480	134319	88210	4638	5778
Corrected	43772	149381	89569	2749	689

Figure 10. Boxplot of blue marlin lengths by year from the Hawaii-based longline observer dataset. The number of fish measured in each year are listed below the boxplot for the year.

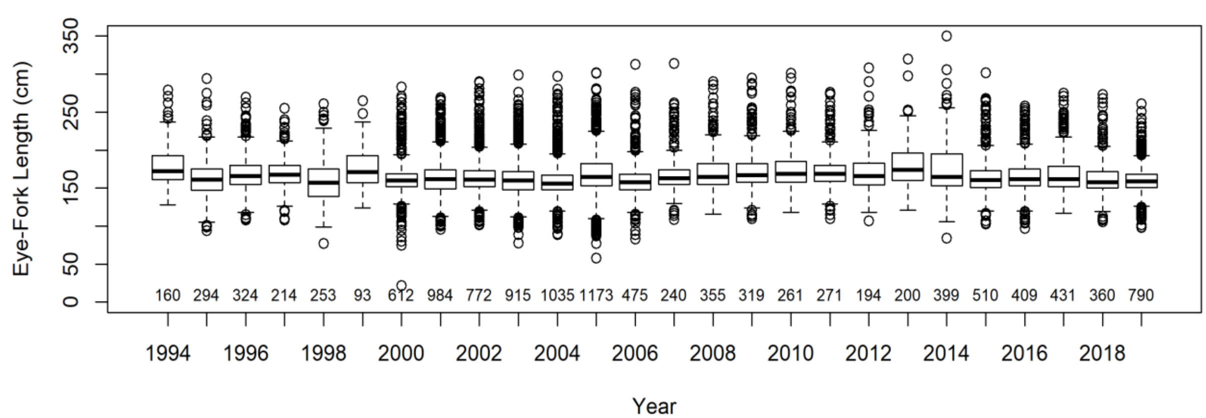


Figure 11. Histogram of all blue marlin lengths available in the Hawaii-based longline observer dataset.

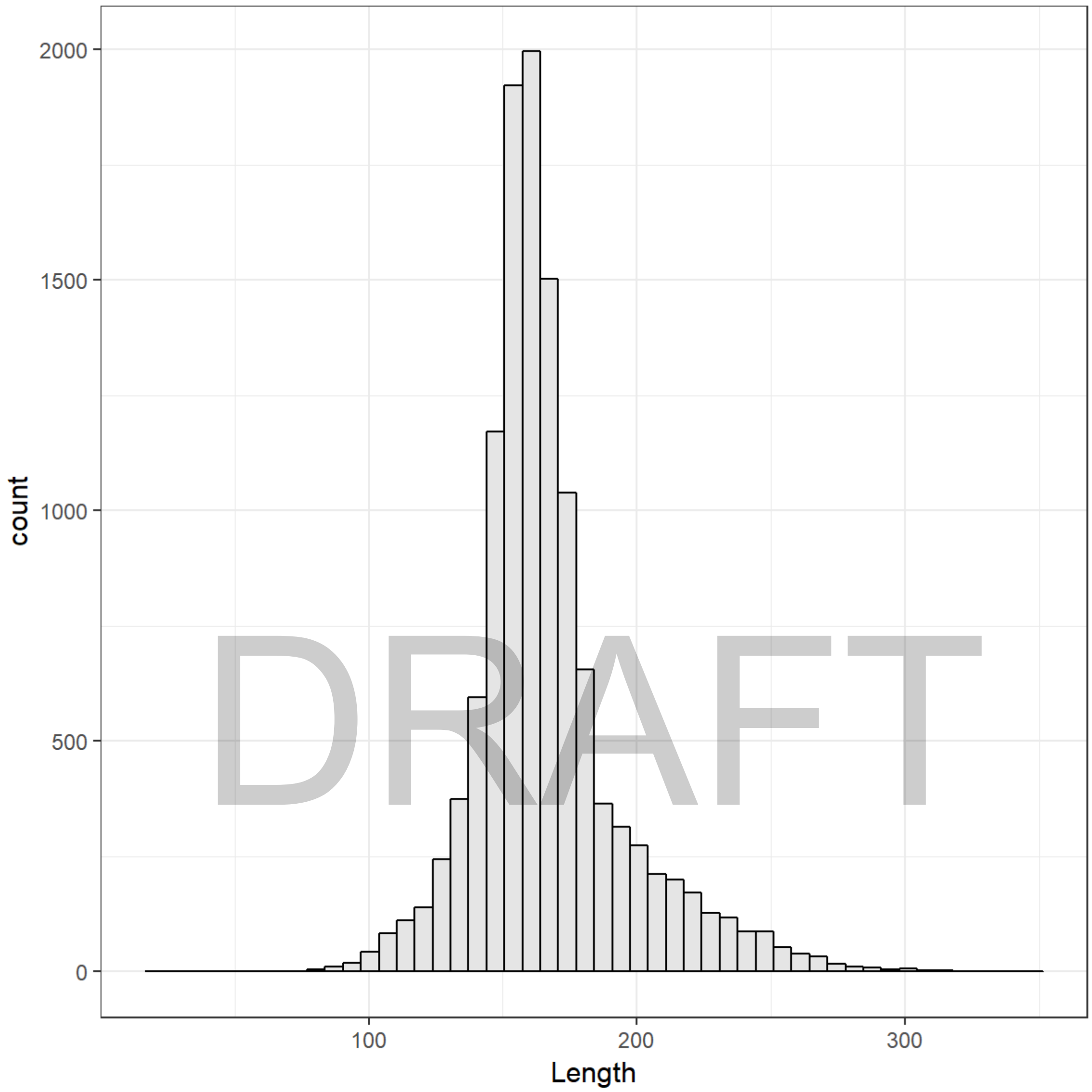


Figure 12. Histograms of lengths of blue marlin available in the Hawaii-based longline observer dataset by quarter (Quarter 1 top left, quarter 2 top right, quarter 3 bottom left, quarter 4 bottom right). Red dashed line indicates length at 50% maturity for females (177cm).

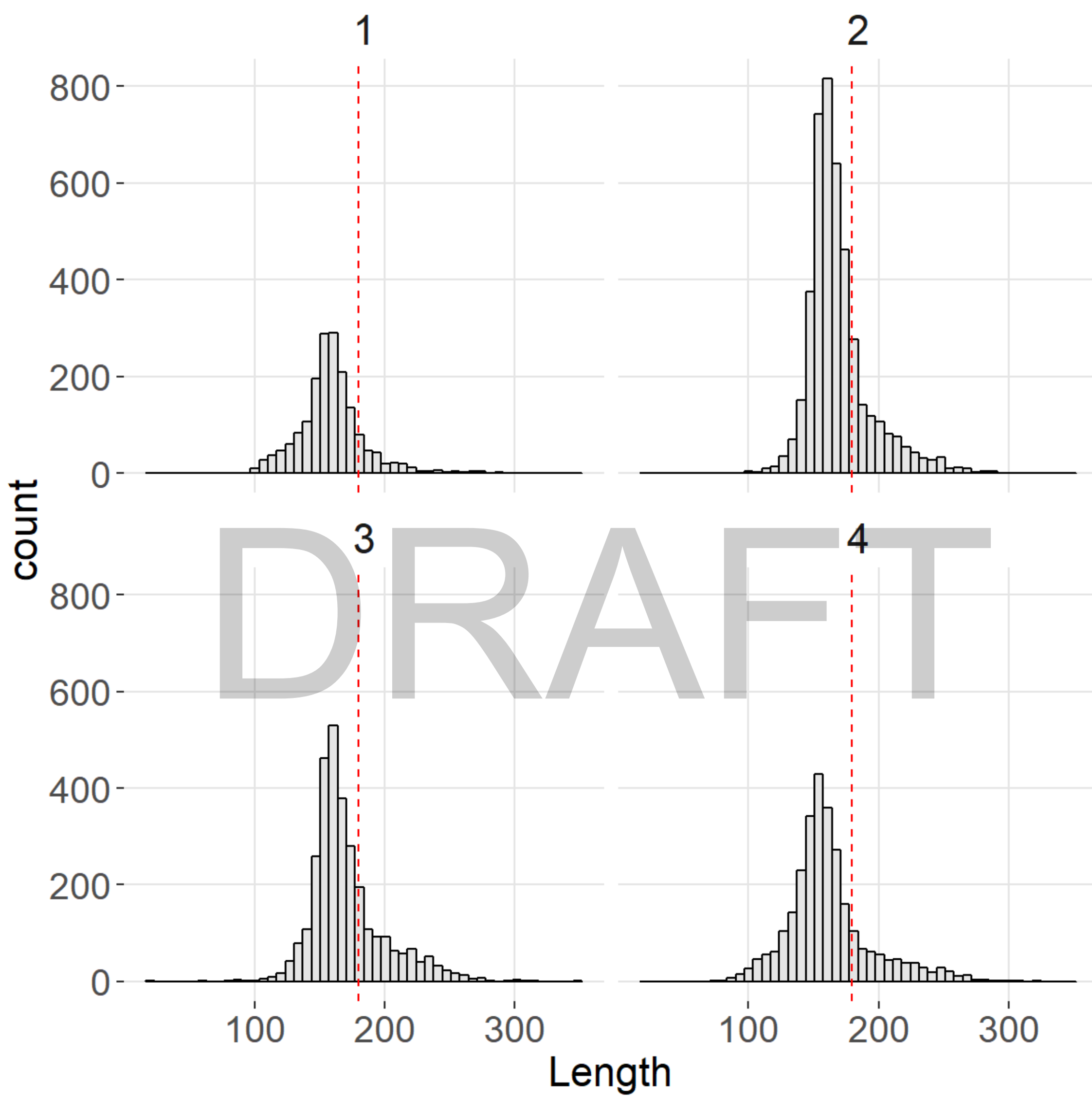


Figure 13. Density plot of blue marlin lengths (in cm) available in the Hawaii-based longline observer dataset. Grey shading indicates fish caught in the deep-set sector, red shading indicates fish caught in the shallow-set sector.

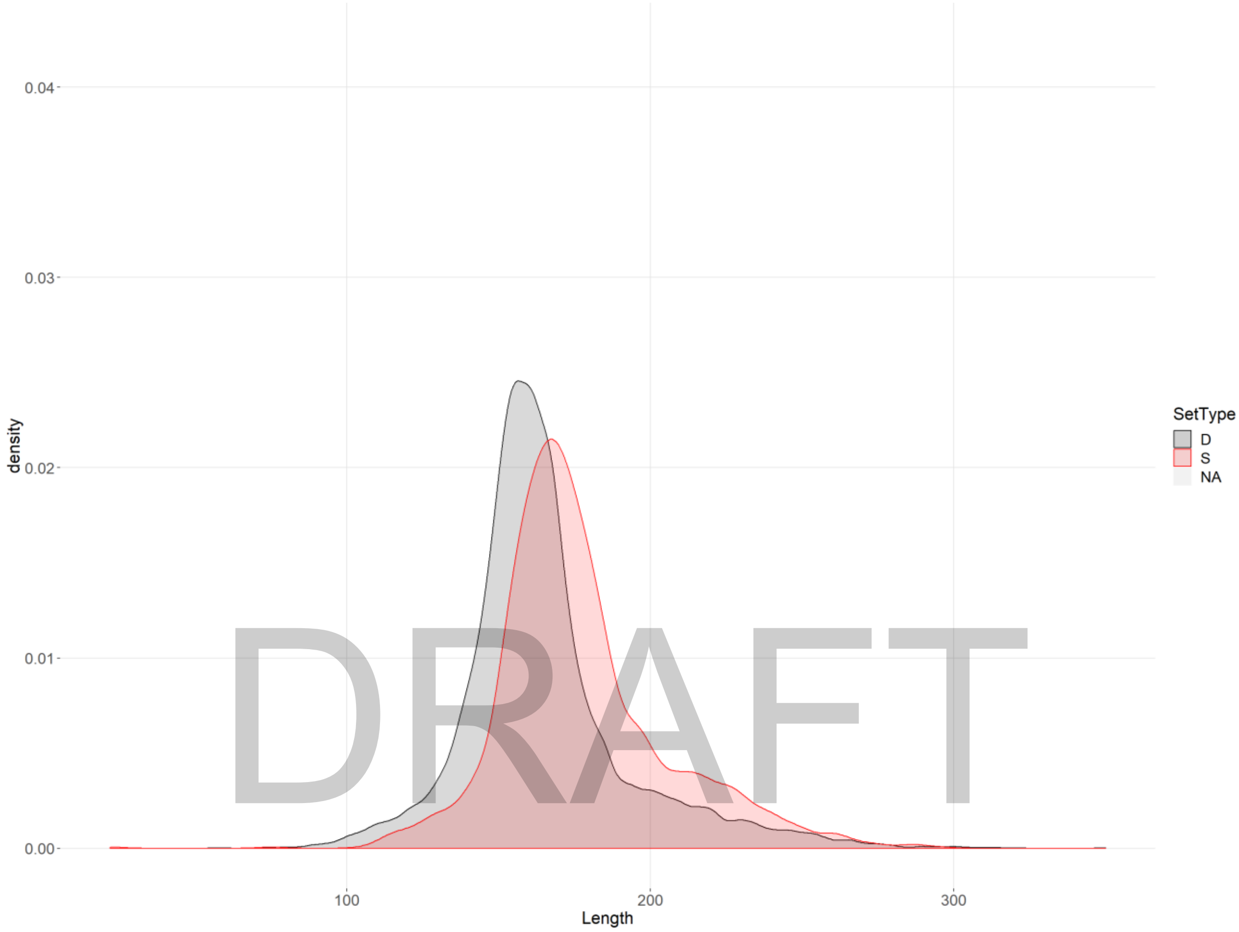


Figure 14. Density plot of blue marlin lengths (in cm) available in the Hawaii-based longline observer dataset. Grey shading indicates females, red shading males, and blue indicates fish with unidentified genders.

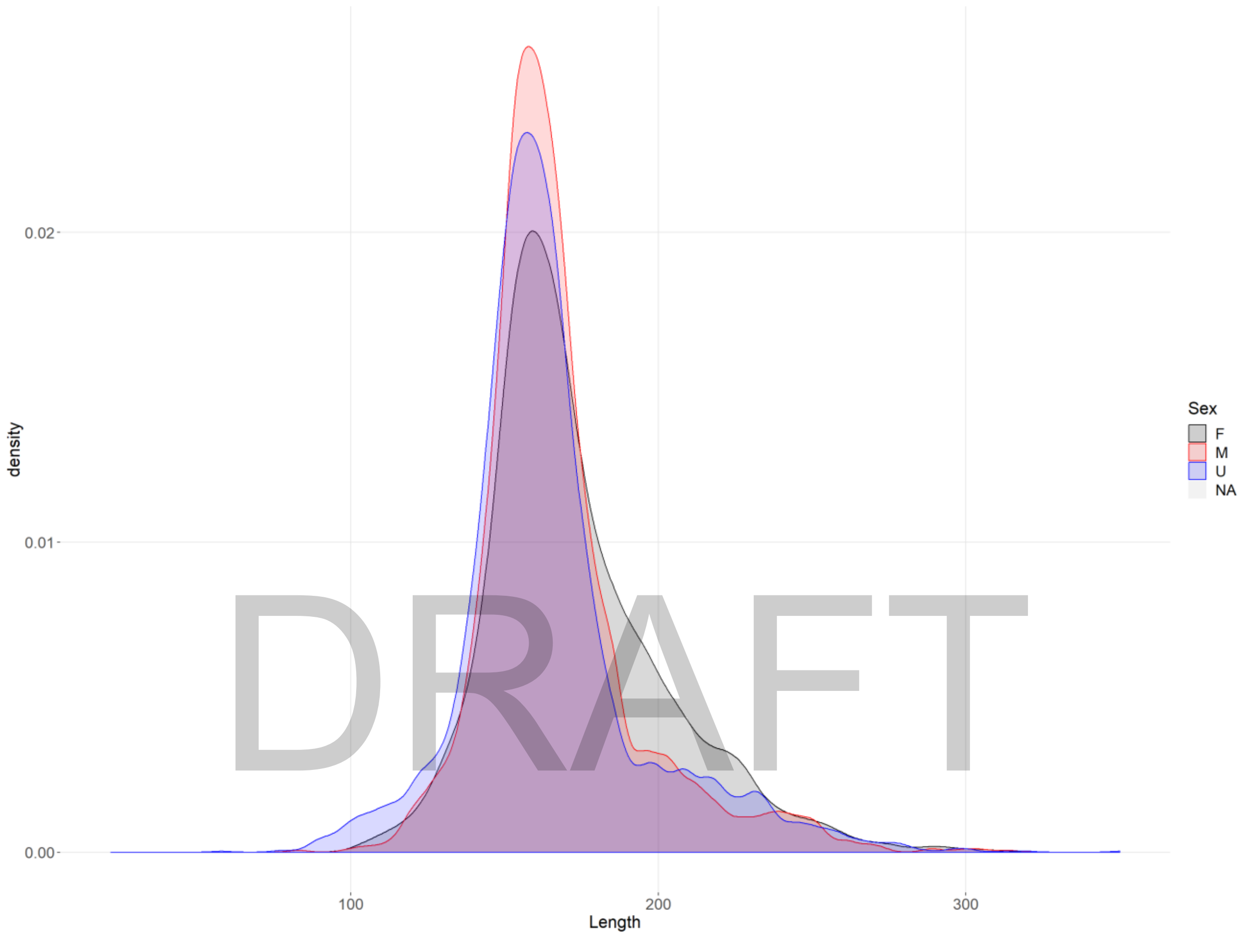


Figure 15. Mean length (EFL in cm) by 5x5° squares of blue marlin caught in the Hawaii-based longline fishery. Squares with fewer than 3 vessels with measured fish were removed for confidentiality.

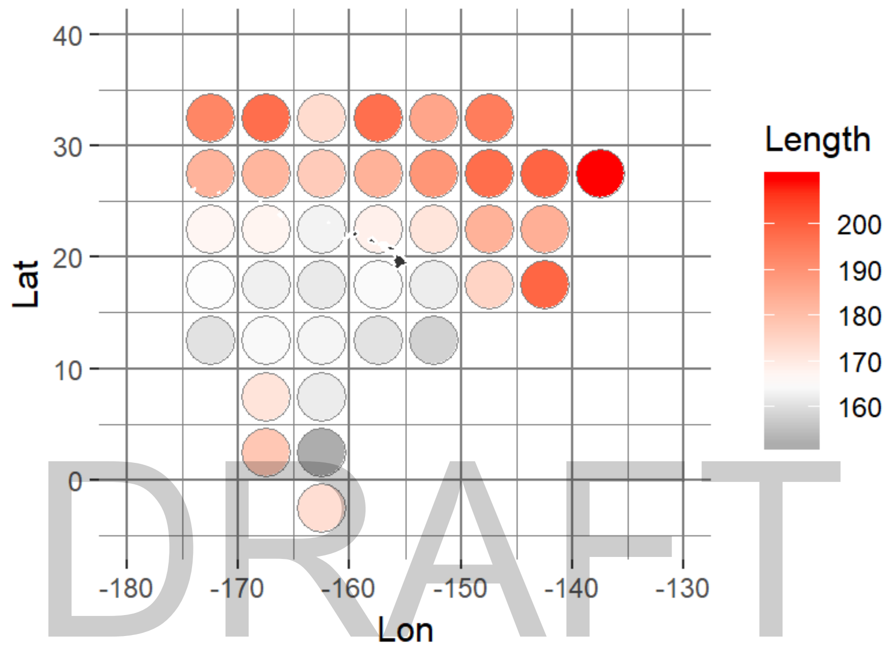


Figure 16. Mean lengths (EFL in cm) by quarter of blue marlin caught in the Hawaii-based longline fishery (quarter 1 top left, quarter 2 top right, quarter 3 bottom left, quarter 4 bottom right). Squares with fewer than three vessels with measured fish were removed for confidentiality.